Kang-Kang Shao

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EDUCATION BACKGROUND

Lanzhou University **Lanzhou, China**
Master in High Energy Physics Theory **09.2022-06.2025**
Major Courses (UCAS): Gauge Field Theory, Lie Group and Lie Algebra, Particle Physics II.
Major Courses (HUST): Quantum Field Theory, Group Theory, Advanced Quantum Mechanics.
Major Research (HUST&LZU): The semi-leptonic inclusive charm decay
GPA: 3.94/5

Lanzhou University **Lanzhou, China**
Bachelor of nuclear physics **09.2018-06.2022**
Rank: 19/52

RESEARCH EXPERIENCE

< The semi-leptonic inclusive charm decay > (2023/08~Now)

- **Theoretically :** Based on Heavy Quark Effective Theory (HQET)and Operator Product Expansion (OPE), we give the analytic result of decay width, high-order normalization electronic energy moment, which contains the NLO corrections of high dimension operator matrix. (**Independently conducted numerical verification of the analytical results simultaneously**)
- **Phenomenologically:** Based on Bayesian analysis, we explore the possibility of using electron energy moments from CLEO measurements of charmed hadron semileptonic inclusive decays to extract non-perturbative parameters.(In collaboration with Huang Chun, who is studying at Washington University in St. Louis, USA)
- **Experimentally :** Based on the data of CLEO and BESIII, we extract the electronic energy normalization momentums of D to X e^+ process, which allow us more systematic research. (In collaboration with the BESIII Collaboration)

< Inclusive weak-annihilation decays and lifetimes of beauty-charmed baryons> (2022/04~2022/08)

- The Ξ_{bc} has not been discovered so far. Using Heavy Di-quark Effective Theory (HDET) and OPE, we have provided the relationship between the decay width of the inclusive decay channel Ξ_{bc} to X_{cs} and the mixing angle. (Undergraduate thesis, Independently derived partial results)

PUBLICATIONS AND PRESENTATION

✓ **HFCPV2024(National level): "Charm Semi-Leptonic Inclusive Decay", Plenary Presentation.**
✓ Guo-He Yang En-Pei Liang, Qin Qin ,and **Kang-Kang Shao**. Inclusive weak-annihilation decays and lifetimes of beauty-charmed baryons [**arXiv:2208.06834** [hep-ph]] **(Phys.Rev.D)**.

SKILLS

Languages: Mandarin(Native), English (IELTS: Preparing)
Computer Skills: Mathmatica (main), Latex(main), Python, Microsoft Office (Word, Excel, and PowerPoint), Origin and so on.

SCHOLARSHIP & AWARDS

Lanzhou University Outstanding Student Scholarship (3 times)	school level	2022-2024
Lanzhou University Outstanding Student Scholarship (3 times)	school level	2019-2021
"Think Tank · Star of Scientific Innovation" Scholarship	school	2021-2022
National Third Prize in the 5th National University "Nuclear + X" Innovation Competition	National Level	2021-2022

Conferences Attended

1. The 21st National Workshop on Heavy Flavor Physics and CP Violation (HFCPV-2024) **Plenary Presentation**
2. The 3rd Joint Workshop on Hadron and Heavy Flavor Physics: Theory and Experiment Volunteer
3. 2023 Summer School on QCD and Medium-to-High Energy Nuclear Physics. Student
4. The Fourth Symposium on Heavy Flavor Physics and Quantum Chromodynamics **Assist in** organizing meetings